

Answer Sheet - Grade 9 Physical Science - Forces and Motion - Short Test

Q1. An object is moving at a steady speed in a straight line. The net force on it is zero.

What will the object do next?

☐ A ☐ B ☒ C ☐ D

Q2. Which unit is used to measure force?

☐ A ☒ B ☐ C ☐ D

Q3. A 2 kg trolley accelerates at 3 m/s squared. What is the net force on it?

☐ A ☐ B ☒ C ☐ D

Q4. A book rests on a table. Using forces, explain why the book is not accelerating.

Gravity pulls it down and the table pushes up with the same force, so the forces are balanced and the net force is zero. No net force means no acceleration.

Q5. A cyclist rides at a steady speed along a flat, straight road. (a) The forward force from the pedals is ... the total resistive force. Mark one letter.

☐ A ☒ B ☐ C

Q5. (b) Explain your choice in (a).

Steady speed means no acceleration so the net force is zero and the forward force equals the resistance.